DC1 universal board: population summary

One 100×100 mm board, four builds (a–d). Each build fits only the parts listed for its letter; every other resistor footprint is left open. Never fit a 0 Ω link in a DNP position. 532 resistor sites total: 512 input sites (full 64 inputs × 4 neurons × POS/INV matrix, every cell has a footprint) + 20 central bias/balance/feedback sites. 112 of these are future-proof additions, DNP on every card.

Card	Neurons	Resistors fitted	5% (weights, bias)	1% (balance, feedback)	DNP footprints	of which new (DNP+)
DC1a	N0–N3	191	183	8	341	112
DC1b	N4–N7	230	222	8	302	112
DC1c	N8–N11	181	173	8	351	112
DC1d	N12–N15	141	133	8	391	112

Fixed parts (fit on every card)

Ref	Part	Cards
U1	TL074 quad op-amp, SOIC-14	all cards
C_U1P, C_U1N	0.1 µF 50 V 0805 decoupling	all cards
J_STRADDLE_EDGE	76-pin edge straddle	all cards

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). **DNP+** on purple = future-proof site added in the 532-site layout, open on all four cards.

Board layout: IN0–IN31 bank on the left half, IN32–IN63 on the right, one row per input, eight columns per bank (POS/INV for local N0–N3). The 20 central resistors sit in two columns at x = 47 / 52.6 mm, y = 44–94 mm (bias on +5V_REF to every POS/INV node, balance on GND to every node, feedback to V_SUM). Reference numbers in each cell match the silkscreen.

DC1a · weights IN0–IN31 (left bank) · neurons N0–N3

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). **DNP+** on purple = future-proof site added in the 532-site layout, open on all four cards.

IN	POS N0 (POS_N0)	INV N0 (INV_N0)	POS N1 (POS_N1)	INV N1 (INV_N1)	POS N2 (POS_N2)	INV N2 (INV_N2)	POS N3 (POS_N3)	INV N3 (INV_N3)
IN0	R001 DNP	R002 30k	R003 DNP	R004 DNP	R005 DNP	R006 DNP	R007 DNP	R008 47k
IN1	R009 390k	R010 DNP	R421 DNP+	R011 DNP	R422 DNP+	R012 110k	R013 2M	R014 DNP
IN2	R015 160k	R016 DNP	R423 DNP+	R017 DNP	R018 1.5M	R019 DNP	R020 120k	R424 DNP+
IN3	R021 180k	R022 DNP	R023 DNP	R024 DNP	R025 DNP	R026 680k	R027 160k	R425 DNP+
IN4	R028 DNP	R029 130k	R030 DNP	R031 DNP	R426 DNP+	R032 510k	R033 150k	R427 DNP+
IN5	R034 DNP	R035 360k	R036 DNP	R037 DNP	R038 110k	R039 DNP	R040 240k	R428 DNP+
IN6	R041 DNP	R042 270k	R429 DNP+	R043 DNP	R044 200k	R045 DNP	R046 430k	R430 DNP+
IN7	R431 DNP+	R047 DNP	R432 DNP+	R048 DNP	R049 DNP	R050 150k	R051 DNP	R052 180k
IN8	R053 360k	R054 DNP	R055 DNP	R056 82k	R433 DNP+	R057 DNP	R058 DNP	R059 130k
IN9	R060 200k	R061 DNP	R062 360k	R063 DNP	R064 240k	R065 DNP	R066 470k	R434 DNP+
IN10	R067 390k	R068 DNP	R435 DNP+	R069 DNP	R436 DNP+	R070 560k	R437 DNP+	R071 620k
IN11	R072 DNP	R073 DNP	R074 DNP	R075 DNP	R076 680k	R077 DNP	R438 DNP+	R439 DNP+
IN12	R078 DNP	R079 160k	R080 DNP	R440 DNP+	R441 DNP+	R081 DNP	R442 DNP+	R082 DNP
IN13	R083 DNP	R084 220k	R443 DNP+	R085 DNP	R444 DNP+	R086 360k	R087 3.3M	R088 DNP
IN14	R089 DNP	R090 270k	R091 DNP	R092 DNP	R445 DNP+	R093 330k	R446 DNP+	R094 1.1M
IN15	R447 DNP+	R095 150k	R448 DNP+	R096 DNP	R449 DNP+	R097 DNP	R450 DNP+	R098 300k
IN16	R099 DNP	R100 DNP	R101 DNP	R102 DNP	R103 DNP	R104 160k	R105 DNP	R106 47k
IN17	R107 DNP	R108 DNP	R109 DNP	R110 DNP	R111 DNP	R112 220k	R113 DNP	R114 56k
IN18	R115 DNP	R116 910k	R451 DNP+	R117 DNP	R118 DNP	R119 160k	R120 DNP	R121 91k
IN19	R122 82k	R452 DNP+	R123 DNP	R124 DNP	R125 DNP	R126 220k	R453 DNP+	R127 130k
IN20	R128 430k	R454 DNP+	R455 DNP+	R129 DNP	R130 DNP	R131 180k	R456 DNP+	R132 2.2M
IN21	R457 DNP+	R133 120k	R458 DNP+	R134 DNP	R135 DNP	R136 100k	R459 DNP+	R137 430k
IN22	R460 DNP+	R138 130k	R461 DNP+	R139 DNP	R140 DNP	R141 91k	R462 DNP+	R142 2M
IN23	R143 DNP	R144 180k	R463 DNP+	R145 DNP	R146 DNP	R147 47k	R464 DNP+	R148 560k
IN24	R465 DNP+	R149 150k	R150 DNP	R151 DNP	R152 DNP	R153 240k	R154 75k	R155 DNP
IN25	R156 430k	R157 DNP	R158 DNP	R159 DNP	R160 DNP	R161 DNP	R162 75k	R163 DNP
IN26	R164 390k	R165 DNP	R466 DNP+	R166 DNP	R467 DNP+	R167 270k	R168 150k	R169 DNP
IN27	R170 160k	R171 DNP	R468 DNP+	R172 DNP	R469 DNP+	R173 330k	R174 150k	R175 DNP
IN28	R176 DNP	R177 300k	R470 DNP+	R178 DNP	R179 DNP	R180 680k	R181 620k	R471 DNP+
IN29	R182 DNP	R183 240k	R472 DNP+	R184 DNP	R185 DNP	R186 200k	R187 330k	R473 DNP+
IN30	R474 DNP+	R188 1.6M	R189 DNP	R190 DNP	R191 DNP	R192 330k	R193 560k	R475 DNP+
IN31	R194 620k	R195 DNP	R196 DNP	R197 75k	R198 220k	R476 DNP+	R199 180k	R477 DNP+

DC1a · weights IN32–IN63 (right bank) · neurons N0–N3

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). **DNP+** on purple = future-proof site added in the 532-site layout, open on all four cards.

IN	POS N0 (POS_N0)	INV N0 (INV_N0)	POS N1 (POS_N1)	INV N1 (INV_N1)	POS N2 (POS_N2)	INV N2 (INV_N2)	POS N3 (POS_N3)	INV N3 (INV_N3)
IN32	R200 360k	R478 DNP+	R201 DNP	R202 91k	R203 910k	R204 DNP	R205 27k	R206 DNP
IN33	R207 1M	R208 DNP	R209 110k	R479 DNP+	R210 180k	R211 DNP	R212 91k	R480 DNP+
IN34	R213 DNP	R214 1.3M	R215 DNP	R481 DNP+	R216 180k	R217 DNP	R218 390k	R219 DNP
IN35	R220 DNP	R221 680k	R222 DNP	R223 200k	R224 220k	R225 DNP	R482 DNP+	R226 330k
IN36	R227 DNP	R228 470k	R483 DNP+	R229 DNP	R230 270k	R231 DNP	R232 DNP	R233 DNP
IN37	R234 DNP	R235 1.2M	R236 DNP	R237 470k	R238 DNP	R239 360k	R484 DNP+	R240 DNP
IN38	R241 330k	R485 DNP+	R242 DNP	R243 DNP	R244 DNP	R245 430k	R246 300k	R486 DNP+
IN39	R247 110k	R248 DNP	R249 DNP	R250 DNP	R251 DNP	R252 75k	R253 DNP	R254 150k
IN40	R255 DNP	R256 510k	R257 DNP	R487 DNP+	R258 DNP	R259 DNP	R488 DNP+	R260 43k
IN41	R261 330k	R262 DNP	R263 110k	R489 DNP+	R264 DNP	R490 DNP+	R491 DNP+	R265 240k
IN42	R266 390k	R267 DNP	R268 DNP	R492 DNP+	R269 150k	R493 DNP+	R270 DNP	R494 DNP+
IN43	R271 390k	R272 DNP	R273 DNP	R495 DNP+	R274 91k	R275 DNP	R496 DNP+	R276 750k
IN44	R277 1.2M	R278 DNP	R279 DNP	R497 DNP+	R280 330k	R281 DNP	R498 DNP+	R282 2.4M
IN45	R283 DNP	R284 DNP	R285 390k	R286 DNP	R287 DNP	R288 DNP	R499 DNP+	R289 2M
IN46	R290 DNP	R500 DNP+	R291 DNP	R501 DNP+	R292 DNP	R293 300k	R294 DNP	R295 1M
IN47	R296 68k	R297 DNP	R298 DNP	R299 DNP	R300 DNP	R301 110k	R302 DNP	R303 110k
IN48	R304 DNP	R305 DNP	R502 DNP+	R306 DNP	R307 DNP	R308 240k	R309 470k	R503 DNP+
IN49	R310 240k	R311 DNP	R312 110k	R504 DNP+	R313 270k	R505 DNP+	R506 DNP+	R314 470k
IN50	R315 330k	R316 DNP	R507 DNP+	R508 DNP+	R317 DNP	R318 680k	R509 DNP+	R319 910k
IN51	R320 560k	R321 DNP	R322 DNP	R323 680k	R324 DNP	R325 330k	R326 470k	R327 DNP
IN52	R328 560k	R329 DNP	R330 DNP	R331 1.2M	R332 3.3M	R510 DNP+	R333 DNP	R334 DNP
IN53	R335 560k	R336 DNP	R337 200k	R338 DNP	R339 390k	R340 DNP	R511 DNP+	R341 1.1M
IN54	R512 DNP+	R342 430k	R513 DNP+	R343 DNP	R344 270k	R345 DNP	R346 430k	R514 DNP+
IN55	R347 DNP	R348 82k	R349 DNP	R350 DNP	R351 DNP	R352 200k	R353 DNP	R354 91k
IN56	R515 DNP+	R355 62k	R356 DNP	R357 DNP	R358 DNP	R359 DNP	R516 DNP+	R360 75k
IN57	R361 DNP	R362 DNP	R363 DNP	R364 DNP	R365 220k	R366 DNP	R517 DNP+	R367 470k
IN58	R518 DNP+	R368 DNP	R369 DNP	R370 DNP	R519 DNP+	R371 390k	R372 DNP	R373 DNP
IN59	R520 DNP+	R374 360k	R375 DNP	R521 DNP+	R522 DNP+	R376 200k	R377 200k	R378 DNP
IN60	R523 DNP+	R379 1.1M	R380 DNP	R381 130k	R382 DNP	R383 220k	R384 330k	R385 DNP
IN61	R386 240k	R387 DNP	R388 DNP	R389 DNP	R390 130k	R391 DNP	R524 DNP+	R392 330k
IN62	R525 DNP+	R393 200k	R394 DNP	R395 120k	R396 100k	R526 DNP+	R527 DNP+	R397 110k
IN63	R528 DNP+	R398 DNP	R399 DNP	R400 DNP	R401 56k	R402 DNP	R403 100k	R404 DNP

DC1a · central resistors (feedback, bias, balance)

x = 47 / 52.6 mm, y = 44–94 mm, grouped by net pair. Balance and feedback are 1% (blue); bias is 5%. Red and purple rows stay open.

Ref	X mm	Y mm	Pad 1	Pad 2	Role	Neuron	Value	Tol	Original ref
R529	47	44	+5V_REF	POS_N0	bias	N0	DNP+ (new)		
R405	47	59	+5V_REF	INV_N0	bias	N0	**6.2M**	5%	Rb_N0
R530	52.6	44	+5V_REF	POS_N1	bias	N1	DNP+ (new)		
R406	52.6	59	+5V_REF	INV_N1	bias	N1	DNP (open)		
R409	47	69	+5V_REF	POS_N2	bias	N2	DNP (open)		
R407	47	64	+5V_REF	INV_N2	bias	N2	**1.1M**	5%	Rb_N2
R531	47	49	+5V_REF	POS_N3	bias	N3	DNP+ (new)		
R408	52.6	64	+5V_REF	INV_N3	bias	N3	DNP (open)		
R413	47	79	GND	POS_N0	balance	N0	**18.2k**	1%	R_BAL_1_0
R532	52.6	49	GND	INV_N0	balance	N0	DNP+ (new)		
R414	52.6	79	GND	POS_N1	balance	N1	**29.4k**	1%	R_BAL_1_1
R410	52.6	69	GND	INV_N1	balance	N1	DNP (open)		
R415	47	84	GND	POS_N2	balance	N2	**14.7k**	1%	R_BAL_1_2
R411	47	74	GND	INV_N2	balance	N2	DNP (open)		
R416	52.6	84	GND	POS_N3	balance	N3	**22.6k**	1%	R_BAL_1_3
R412	52.6	74	GND	INV_N3	balance	N3	DNP (open)		
R417	47	89	INV_N0	V_SUM_0	feedback	N0	**100k**	1%	Rf_N0
R418	52.6	89	INV_N1	V_SUM_1	feedback	N1	**100k**	1%	Rf_N1
R419	47	94	INV_N2	V_SUM_2	feedback	N2	**100k**	1%	Rf_N2
R420	52.6	94	INV_N3	V_SUM_3	feedback	N3	**100k**	1%	Rf_N3

Also fit the fixed parts: U1 TL074, C_U1P and C_U1N 0.1 µF 50 V 0805, J_STRADDLE_EDGE.

DC1b · weights IN0–IN31 (left bank) · neurons N4–N7

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). **DNP+** on purple = future-proof site added in the 532-site layout, open on all four cards.

IN	POS N4 (POS_N0)	INV N4 (INV_N0)	POS N5 (POS_N1)	INV N5 (INV_N1)	POS N6 (POS_N2)	INV N6 (INV_N2)	POS N7 (POS_N3)	INV N7 (INV_N3)
IN0	R001 DNP	R002 51k	R003 130k	R004 DNP	R005 DNP	R006 75k	R007 75k	R008 DNP
IN1	R009 DNP	R010 200k	R421 DNP+	R011 150k	R422 DNP+	R012 240k	R013 91k	R014 DNP
IN2	R015 DNP	R016 1.3M	R423 DNP+	R017 200k	R018 DNP	R019 DNP	R020 330k	R424 DNP+
IN3	R021 430k	R022 DNP	R023 DNP	R024 150k	R025 680k	R026 DNP	R027 330k	R425 DNP+
IN4	R028 DNP	R029 270k	R030 DNP	R031 150k	R426 DNP+	R032 110k	R033 110k	R427 DNP+
IN5	R034 DNP	R035 510k	R036 DNP	R037 300k	R038 DNP	R039 82k	R040 130k	R428 DNP+
IN6	R041 DNP	R042 100k	R429 DNP+	R043 160k	R044 DNP	R045 56k	R046 82k	R430 DNP+
IN7	R431 DNP+	R047 DNP	R432 DNP+	R048 33k	R049 DNP	R050 47k	R051 390k	R052 DNP
IN8	R053 DNP	R054 100k	R055 470k	R056 DNP	R433 DNP+	R057 110k	R058 150k	R059 DNP
IN9	R060 270k	R061 DNP	R062 DNP	R063 DNP	R064 DNP	R065 1M	R066 DNP	R434 DNP+
IN10	R067 DNP	R068 1.8M	R435 DNP+	R069 160k	R436 DNP+	R070 1.6M	R437 DNP+	R071 DNP
IN11	R072 DNP	R073 330k	R074 DNP	R075 300k	R076 DNP	R077 430k	R438 DNP+	R439 DNP+
IN12	R078 DNP	R079 820k	R080 DNP	R440 DNP+	R441 DNP+	R081 270k	R442 DNP+	R082 1.1M
IN13	R083 DNP	R084 DNP	R443 DNP+	R085 DNP	R444 DNP+	R086 200k	R087 DNP	R088 300k
IN14	R089 510k	R090 DNP	R091 1.8M	R092 DNP	R445 DNP+	R093 130k	R446 DNP+	R094 360k
IN15	R447 DNP+	R095 160k	R448 DNP+	R096 160k	R449 DNP+	R097 33k	R450 DNP+	R098 75k
IN16	R099 DNP	R100 DNP	R101 DNP	R102 220k	R103 220k	R104 DNP	R105 510k	R106 DNP
IN17	R107 DNP	R108 DNP	R109 DNP	R110 330k	R111 DNP	R112 DNP	R113 560k	R114 DNP
IN18	R115 DNP	R116 330k	R451 DNP+	R117 DNP	R118 DNP	R119 DNP	R120 910k	R121 DNP
IN19	R122 DNP	R452 DNP+	R123 DNP	R124 360k	R125 160k	R126 DNP	R453 DNP+	R127 DNP
IN20	R128 240k	R454 DNP+	R455 DNP+	R129 430k	R130 82k	R131 DNP	R456 DNP+	R132 DNP
IN21	R457 DNP+	R133 820k	R458 DNP+	R134 DNP	R135 75k	R136 DNP	R459 DNP+	R137 270k
IN22	R460 DNP+	R138 680k	R461 DNP+	R139 470k	R140 56k	R141 DNP	R462 DNP+	R142 75k
IN23	R143 DNP	R144 DNP	R463 DNP+	R145 91k	R146 51k	R147 DNP	R464 DNP+	R148 24k
IN24	R465 DNP+	R149 91k	R150 300k	R151 DNP	R152 DNP	R153 DNP	R154 DNP	R155 110k
IN25	R156 DNP	R157 300k	R158 360k	R159 DNP	R160 DNP	R161 DNP	R162 DNP	R163 510k
IN26	R164 DNP	R165 750k	R466 DNP+	R166 510k	R467 DNP+	R167 DNP	R168 DNP	R169 1.5M
IN27	R170 160k	R171 DNP	R468 DNP+	R172 120k	R469 DNP+	R173 DNP	R174 DNP	R175 2M
IN28	R176 220k	R177 DNP	R470 DNP+	R178 180k	R179 DNP	R180 200k	R181 220k	R471 DNP+
IN29	R182 300k	R183 DNP	R472 DNP+	R184 1.3M	R185 DNP	R186 330k	R187 62k	R473 DNP+
IN30	R474 DNP+	R188 220k	R189 430k	R190 DNP	R191 150k	R192 DNP	R193 20k	R475 DNP+
IN31	R194 DNP	R195 82k	R196 100k	R197 DNP	R198 68k	R476 DNP+	R199 43k	R477 DNP+

DC1b · weights IN32–IN63 (right bank) · neurons N4–N7

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). **DNP+** on purple = future-proof site added in the 532-site layout, open on all four cards.

IN	POS N4 (POS_N0)	INV N4 (INV_N0)	POS N5 (POS_N1)	INV N5 (INV_N1)	POS N6 (POS_N2)	INV N6 (INV_N2)	POS N7 (POS_N3)	INV N7 (INV_N3)
IN32	R200 82k	R478 DNP+	R201 91k	R202 DNP	R203 560k	R204 DNP	R205 DNP	R206 160k
IN33	R207 360k	R208 DNP	R209 91k	R479 DNP+	R210 120k	R211 DNP	R212 680k	R480 DNP+
IN34	R213 390k	R214 DNP	R215 180k	R481 DNP+	R216 680k	R217 DNP	R218 DNP	R219 1.1M
IN35	R220 200k	R221 DNP	R222 DNP	R223 300k	R224 DNP	R225 390k	R482 DNP+	R226 430k
IN36	R227 150k	R228 DNP	R483 DNP+	R229 470k	R230 DNP	R231 560k	R232 DNP	R233 750k
IN37	R234 470k	R235 DNP	R236 620k	R237 DNP	R238 DNP	R239 1M	R484 DNP+	R240 270k
IN38	R241 130k	R485 DNP+	R242 330k	R243 DNP	R244 DNP	R245 330k	R246 220k	R486 DNP+
IN39	R247 330k	R248 DNP	R249 DNP	R250 300k	R251 DNP	R252 68k	R253 27k	R254 DNP
IN40	R255 51k	R256 DNP	R257 120k	R487 DNP+	R258 DNP	R259 110k	R488 DNP+	R260 110k
IN41	R261 51k	R262 DNP	R263 150k	R489 DNP+	R264 470k	R490 DNP+	R491 DNP+	R265 DNP
IN42	R266 82k	R267 DNP	R268 82k	R492 DNP+	R269 470k	R493 DNP+	R270 510k	R494 DNP+
IN43	R271 100k	R272 DNP	R273 110k	R495 DNP+	R274 330k	R275 DNP	R496 DNP+	R276 680k
IN44	R277 390k	R278 DNP	R279 180k	R497 DNP+	R280 510k	R281 DNP	R498 DNP+	R282 300k
IN45	R283 300k	R284 DNP	R285 240k	R286 DNP	R287 DNP	R288 240k	R499 DNP+	R289 510k
IN46	R290 200k	R500 DNP+	R291 820k	R501 DNP+	R292 DNP	R293 300k	R294 680k	R295 DNP
IN47	R296 DNP	R297 360k	R298 DNP	R299 82k	R300 DNP	R301 200k	R302 82k	R303 DNP
IN48	R304 DNP	R305 91k	R502 DNP+	R306 150k	R307 510k	R308 DNP	R309 DNP	R503 DNP+
IN49	R310 DNP	R311 200k	R312 DNP	R504 DNP+	R313 510k	R505 DNP+	R506 DNP+	R314 1.1M
IN50	R315 DNP	R316 180k	R507 DNP+	R508 DNP+	R317 DNP	R318 470k	R509 DNP+	R319 DNP
IN51	R320 DNP	R321 120k	R322 200k	R323 DNP	R324 DNP	R325 2M	R326 DNP	R327 2.2M
IN52	R328 DNP	R329 200k	R330 750k	R331 DNP	R332 DNP	R510 DNP+	R333 DNP	R334 750k
IN53	R335 DNP	R336 360k	R337 DNP	R338 DNP	R339 300k	R340 DNP	R511 DNP+	R341 DNP
IN54	R512 DNP+	R342 130k	R513 DNP+	R343 510k	R344 DNP	R345 470k	R346 DNP	R514 DNP+
IN55	R347 DNP	R348 150k	R349 DNP	R350 160k	R351 DNP	R352 DNP	R353 91k	R354 DNP
IN56	R515 DNP+	R355 DNP	R356 DNP	R357 39k	R358 160k	R359 DNP	R516 DNP+	R360 DNP
IN57	R361 DNP	R362 200k	R363 DNP	R364 1.5M	R365 560k	R366 DNP	R517 DNP+	R367 160k
IN58	R518 DNP+	R368 180k	R369 DNP	R370 270k	R519 DNP+	R371 1.8M	R372 DNP	R373 820k
IN59	R520 DNP+	R374 330k	R375 750k	R521 DNP+	R522 DNP+	R376 680k	R377 DNP	R378 240k
IN60	R523 DNP+	R379 200k	R380 1.6M	R381 DNP	R382 DNP	R383 200k	R384 DNP	R385 560k
IN61	R386 DNP	R387 120k	R388 DNP	R389 330k	R390 DNP	R391 270k	R524 DNP+	R392 DNP
IN62	R525 DNP+	R393 DNP	R394 1M	R395 DNP	R396 200k	R526 DNP+	R527 DNP+	R397 180k
IN63	R528 DNP+	R398 62k	R399 DNP	R400 75k	R401 DNP	R402 DNP	R403 DNP	R404 51k

DC1b · central resistors (feedback, bias, balance)

x = 47 / 52.6 mm, y = 44–94 mm, grouped by net pair. Balance and feedback are 1% (blue); bias is 5%. Red and purple rows stay open.

Ref	X mm	Y mm	Pad 1	Pad 2	Role	Neuron	Value	Tol	Original ref
R529	47	44	+5V_REF	POS_N0	bias	N4	DNP+ (new)		
R405	47	59	+5V_REF	INV_N0	bias	N4	**330k**	5%	Rb_N4
R530	52.6	44	+5V_REF	POS_N1	bias	N5	DNP+ (new)		
R406	52.6	59	+5V_REF	INV_N1	bias	N5	**330k**	5%	Rb_N5
R409	47	69	+5V_REF	POS_N2	bias	N6	**1.1M**	5%	Rb_N6
R407	47	64	+5V_REF	INV_N2	bias	N6	DNP (open)		
R531	47	49	+5V_REF	POS_N3	bias	N7	DNP+ (new)		
R408	52.6	64	+5V_REF	INV_N3	bias	N7	**390k**	5%	Rb_N7
R413	47	79	GND	POS_N0	balance	N4	**14.3k**	1%	R_BAL_1_4
R532	52.6	49	GND	INV_N0	balance	N4	DNP+ (new)		
R414	52.6	79	GND	POS_N1	balance	N5	**10k**	1%	R_BAL_1_5
R410	52.6	69	GND	INV_N1	balance	N5	DNP (open)		
R415	47	84	GND	POS_N2	balance	N6	**13k**	1%	R_BAL_1_6
R411	47	74	GND	INV_N2	balance	N6	DNP (open)		
R416	52.6	84	GND	POS_N3	balance	N7	DNP (open)		
R412	52.6	74	GND	INV_N3	balance	N7	**16.2k**	1%	R_BAL_1_7
R417	47	89	INV_N0	V_SUM_0	feedback	N4	**100k**	1%	Rf_N4
R418	52.6	89	INV_N1	V_SUM_1	feedback	N5	**100k**	1%	Rf_N5
R419	47	94	INV_N2	V_SUM_2	feedback	N6	**100k**	1%	Rf_N6
R420	52.6	94	INV_N3	V_SUM_3	feedback	N7	**100k**	1%	Rf_N7

Also fit the fixed parts: U1 TL074, C_U1P and C_U1N 0.1 µF 50 V 0805, J_STRADDLE_EDGE.

DC1c · weights IN0–IN31 (left bank) · neurons N8–N11

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). **DNP+** on purple = future-proof site added in the 532-site layout, open on all four cards.

IN	POS N8 (POS_N0)	INV N8 (INV_N0)	POS N9 (POS_N1)	INV N9 (INV_N1)	POS N10 (POS_N2)	INV N10 (INV_N2)	POS N11 (POS_N3)	INV N11 (INV_N3)
IN0	R001 DNP	R002 DNP	R003 DNP	R004 51k	R005 DNP	R006 DNP	R007 DNP	R008 DNP
IN1	R009 DNP	R010 180k	R421 DNP+	R011 33k	R422 DNP+	R012 43k	R013 DNP	R014 DNP
IN2	R015 DNP	R016 270k	R423 DNP+	R017 DNP	R018 DNP	R019 62k	R020 DNP	R424 DNP+
IN3	R021 DNP	R022 1M	R023 62k	R024 DNP	R025 DNP	R026 120k	R027 DNP	R425 DNP+
IN4	R028 DNP	R029 180k	R030 43k	R031 DNP	R426 DNP+	R032 160k	R033 DNP	R427 DNP+
IN5	R034 DNP	R035 360k	R036 75k	R037 DNP	R038 DNP	R039 240k	R040 DNP	R428 DNP+
IN6	R041 DNP	R042 330k	R429 DNP+	R043 110k	R044 DNP	R045 DNP	R046 DNP	R430 DNP+
IN7	R431 DNP+	R047 47k	R432 DNP+	R048 39k	R049 27k	R050 DNP	R051 DNP	R052 DNP
IN8	R053 DNP	R054 75k	R055 DNP	R056 91k	R433 DNP+	R057 39k	R058 DNP	R059 DNP
IN9	R060 DNP	R061 620k	R062 DNP	R063 360k	R064 DNP	R065 100k	R066 DNP	R434 DNP+
IN10	R067 DNP	R068 220k	R435 DNP+	R069 240k	R436 DNP+	R070 91k	R437 DNP+	R071 DNP
IN11	R072 DNP	R073 360k	R074 180k	R075 DNP	R076 DNP	R077 110k	R438 DNP+	R439 DNP+
IN12	R078 DNP	R079 430k	R080 300k	R440 DNP+	R441 DNP+	R081 330k	R442 DNP+	R082 DNP
IN13	R083 DNP	R084 220k	R443 DNP+	R085 180k	R444 DNP+	R086 560k	R087 DNP	R088 DNP
IN14	R089 DNP	R090 510k	R091 DNP	R092 160k	R445 DNP+	R093 DNP	R446 DNP+	R094 DNP
IN15	R447 DNP+	R095 68k	R448 DNP+	R096 110k	R449 DNP+	R097 82k	R450 DNP+	R098 DNP
IN16	R099 DNP	R100 110k	R101 91k	R102 DNP	R103 47k	R104 DNP	R105 DNP	R106 DNP
IN17	R107 DNP	R108 160k	R109 220k	R110 DNP	R111 47k	R112 DNP	R113 DNP	R114 DNP
IN18	R115 DNP	R116 120k	R451 DNP+	R117 160k	R118 62k	R119 DNP	R120 DNP	R121 DNP
IN19	R122 DNP	R452 DNP+	R123 330k	R124 DNP	R125 75k	R126 DNP	R453 DNP+	R127 DNP
IN20	R128 470k	R454 DNP+	R455 DNP+	R129 390k	R130 DNP	R131 DNP	R456 DNP+	R132 DNP
IN21	R457 DNP+	R133 130k	R458 DNP+	R134 240k	R135 DNP	R136 910k	R459 DNP+	R137 DNP
IN22	R460 DNP+	R138 120k	R461 DNP+	R139 DNP	R140 DNP	R141 270k	R462 DNP+	R142 DNP
IN23	R143 DNP	R144 130k	R463 DNP+	R145 DNP	R146 DNP	R147 180k	R464 DNP+	R148 DNP
IN24	R465 DNP+	R149 200k	R150 DNP	R151 130k	R152 56k	R153 DNP	R154 750k	R155 DNP
IN25	R156 DNP	R157 470k	R158 DNP	R159 150k	R160 750k	R161 DNP	R162 DNP	R163 DNP
IN26	R164 DNP	R165 560k	R466 DNP+	R166 270k	R467 DNP+	R167 DNP	R168 DNP	R169 2.2M
IN27	R170 470k	R171 DNP	R468 DNP+	R172 DNP	R469 DNP+	R173 820k	R174 DNP	R175 DNP
IN28	R176 DNP	R177 DNP	R470 DNP+	R178 240k	R179 270k	R180 DNP	R181 180k	R471 DNP+
IN29	R182 DNP	R183 1.2M	R472 DNP+	R184 150k	R185 1.3M	R186 DNP	R187 DNP	R473 DNP+
IN30	R474 DNP+	R188 200k	R189 DNP	R190 130k	R191 DNP	R192 DNP	R193 DNP	R475 DNP+
IN31	R194 DNP	R195 DNP	R196 3M	R197 DNP	R198 DNP	R476 DNP+	R199 DNP	R477 DNP+

DC1c · weights IN32–IN63 (right bank) · neurons N8–N11

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). **DNP+** on purple = future-proof site added in the 532-site layout, open on all four cards.

IN	POS N8 (POS_N0)	INV N8 (INV_N0)	POS N9 (POS_N1)	INV N9 (INV_N1)	POS N10 (POS_N2)	INV N10 (INV_N2)	POS N11 (POS_N3)	INV N11 (INV_N3)
IN32	R200 110k	R478 DNP+	R201 91k	R202 DNP	R203 DNP	R204 47k	R205 910k	R206 DNP
IN33	R207 430k	R208 DNP	R209 270k	R479 DNP+	R210 DNP	R211 330k	R212 DNP	R480 DNP+
IN34	R213 DNP	R214 DNP	R215 160k	R481 DNP+	R216 910k	R217 DNP	R218 DNP	R219 DNP
IN35	R220 DNP	R221 390k	R222 DNP	R223 DNP	R224 DNP	R225 750k	R482 DNP+	R226 DNP
IN36	R227 DNP	R228 750k	R483 DNP+	R229 1.1M	R230 DNP	R231 1M	R232 DNP	R233 DNP
IN37	R234 270k	R235 DNP	R236 DNP	R237 DNP	R238 270k	R239 DNP	R484 DNP+	R240 DNP
IN38	R241 91k	R485 DNP+	R242 DNP	R243 510k	R244 510k	R245 DNP	R246 DNP	R486 DNP+
IN39	R247 240k	R248 DNP	R249 130k	R250 DNP	R251 270k	R252 DNP	R253 DNP	R254 DNP
IN40	R255 DNP	R256 DNP	R257 68k	R487 DNP+	R258 DNP	R259 DNP	R488 DNP+	R260 DNP
IN41	R261 240k	R262 DNP	R263 DNP	R489 DNP+	R264 390k	R490 DNP+	R491 DNP+	R265 DNP
IN42	R266 470k	R267 DNP	R268 300k	R492 DNP+	R269 DNP	R493 DNP+	R270 DNP	R494 DNP+
IN43	R271 300k	R272 DNP	R273 180k	R495 DNP+	R274 560k	R275 DNP	R496 DNP+	R276 DNP
IN44	R277 270k	R278 DNP	R279 270k	R497 DNP+	R280 DNP	R281 DNP	R498 DNP+	R282 DNP
IN45	R283 180k	R284 DNP	R285 270k	R286 DNP	R287 2M	R288 DNP	R499 DNP+	R289 240k
IN46	R290 150k	R500 DNP+	R291 270k	R501 DNP+	R292 430k	R293 DNP	R294 DNP	R295 DNP
IN47	R296 560k	R297 DNP	R298 39k	R299 DNP	R300 DNP	R301 360k	R302 DNP	R303 DNP
IN48	R304 1.1M	R305 DNP	R502 DNP+	R306 DNP	R307 DNP	R308 270k	R309 DNP	R503 DNP+
IN49	R310 DNP	R311 DNP	R312 270k	R504 DNP+	R313 DNP	R505 DNP+	R506 DNP+	R314 DNP
IN50	R315 220k	R316 DNP	R507 DNP+	R508 DNP+	R317 DNP	R318 DNP	R509 DNP+	R319 DNP
IN51	R320 220k	R321 DNP	R322 750k	R323 DNP	R324 DNP	R325 DNP	R326 DNP	R327 DNP
IN52	R328 620k	R329 DNP	R330 DNP	R331 750k	R332 DNP	R510 DNP+	R333 DNP	R334 DNP
IN53	R335 620k	R336 DNP	R337 DNP	R338 820k	R339 DNP	R340 620k	R511 DNP+	R341 300k
IN54	R512 DNP+	R342 DNP	R513 DNP+	R343 820k	R344 DNP	R345 200k	R346 DNP	R514 DNP+
IN55	R347 1.2M	R348 DNP	R349 56k	R350 DNP	R351 1.3M	R352 DNP	R353 DNP	R354 DNP
IN56	R515 DNP+	R355 62k	R356 110k	R357 DNP	R358 DNP	R359 100k	R516 DNP+	R360 DNP
IN57	R361 300k	R362 DNP	R363 130k	R364 DNP	R365 DNP	R366 330k	R517 DNP+	R367 150k
IN58	R518 DNP+	R368 680k	R369 220k	R370 DNP	R519 DNP+	R371 220k	R372 560k	R373 DNP
IN59	R520 DNP+	R374 360k	R375 390k	R521 DNP+	R522 DNP+	R376 270k	R377 300k	R378 DNP
IN60	R523 DNP+	R379 510k	R380 470k	R381 DNP	R382 820k	R383 DNP	R384 110k	R385 DNP
IN61	R386 DNP	R387 300k	R388 68k	R389 DNP	R390 DNP	R391 DNP	R524 DNP+	R392 DNP
IN62	R525 DNP+	R393 DNP	R394 110k	R395 DNP	R396 200k	R526 DNP+	R527 DNP+	R397 150k
IN63	R528 DNP+	R398 240k	R399 160k	R400 DNP	R401 36k	R402 DNP	R403 DNP	R404 560k

DC1c · central resistors (feedback, bias, balance)

x = 47 / 52.6 mm, y = 44–94 mm, grouped by net pair. Balance and feedback are 1% (blue); bias is 5%. Red and purple rows stay open.

Ref	X mm	Y mm	Pad 1	Pad 2	Role	Neuron	Value	Tol	Original ref
R529	47	44	+5V_REF	POS_N0	bias	N8	DNP+ (new)		
R405	47	59	+5V_REF	INV_N0	bias	N8	**820k**	5%	Rb_N8
R530	52.6	44	+5V_REF	POS_N1	bias	N9	DNP+ (new)		
R406	52.6	59	+5V_REF	INV_N1	bias	N9	**270k**	5%	Rb_N9
R409	47	69	+5V_REF	POS_N2	bias	N10	DNP (open)		
R407	47	64	+5V_REF	INV_N2	bias	N10	DNP (open)		
R531	47	49	+5V_REF	POS_N3	bias	N11	DNP+ (new)		
R408	52.6	64	+5V_REF	INV_N3	bias	N11	DNP (open)		
R413	47	79	GND	POS_N0	balance	N8	**8.66k**	1%	R_BAL_1_8
R532	52.6	49	GND	INV_N0	balance	N8	DNP+ (new)		
R414	52.6	79	GND	POS_N1	balance	N9	DNP (open)		
R410	52.6	69	GND	INV_N1	balance	N9	**18.2k**	1%	R_BAL_1_9
R415	47	84	GND	POS_N2	balance	N10	**35.7k**	1%	R_BAL_1_10
R411	47	74	GND	INV_N2	balance	N10	DNP (open)		
R416	52.6	84	GND	POS_N3	balance	N11	**90.9k**	1%	R_BAL_1_11
R412	52.6	74	GND	INV_N3	balance	N11	DNP (open)		
R417	47	89	INV_N0	V_SUM_0	feedback	N8	**100k**	1%	Rf_N8
R418	52.6	89	INV_N1	V_SUM_1	feedback	N9	**100k**	1%	Rf_N9
R419	47	94	INV_N2	V_SUM_2	feedback	N10	**100k**	1%	Rf_N10
R420	52.6	94	INV_N3	V_SUM_3	feedback	N11	**100k**	1%	Rf_N11

Also fit the fixed parts: U1 TL074, C_U1P and C_U1N 0.1 µF 50 V 0805, J_STRADDLE_EDGE.

DC1d · weights IN0–IN31 (left bank) · neurons N12–N15

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). **DNP+** on purple = future-proof site added in the 532-site layout, open on all four cards.

IN	POS N12 (POS_N0)	INV N12 (INV_N0)	POS N13 (POS_N1)	INV N13 (INV_N1)	POS N14 (POS_N2)	INV N14 (INV_N2)	POS N15 (POS_N3)	INV N15 (INV_N3)
IN0	R001 **43k**	R002 DNP	R003 DNP	R004 DNP	R005 **39k**	R006 DNP	R007 DNP	R008 DNP
IN1	R009 **200k**	R010 DNP	R421 DNP+	R011 DNP	R422 DNP+	R012 **91k**	R013 DNP	R014 **220k**
IN2	R015 **390k**	R016 DNP	R423 DNP+	R017 DNP	R018 DNP	R019 **100k**	R020 DNP	R424 DNP+
IN3	R021 **200k**	R022 DNP	R023 DNP	R024 DNP	R025 DNP	R026 **150k**	R027 DNP	R425 DNP+
IN4	R028 **300k**	R029 DNP	R030 DNP	R031 DNP	R426 DNP+	R032 DNP	R033 DNP	R427 DNP+
IN5	R034 **270k**	R035 DNP	R036 DNP	R037 DNP	R038 DNP	R039 **200k**	R040 DNP	R428 DNP+
IN6	R041 **750k**	R042 DNP	R429 DNP+	R043 DNP	R044 DNP	R045 **110k**	R046 DNP	R430 DNP+
IN7	R431 DNP+	R047 **270k**	R432 DNP+	R048 DNP	R049 DNP	R050 DNP	R051 DNP	R052 DNP
IN8	R053 DNP	R054 **160k**	R055 DNP	R056 **430k**	R433 DNP+	R057 **36k**	R058 DNP	R059 **180k**
IN9	R060 **910k**	R061 DNP	R062 DNP	R063 DNP	R064 DNP	R065 **270k**	R066 DNP	R434 DNP+
IN10	R067 **300k**	R068 DNP	R435 DNP+	R069 DNP	R436 DNP+	R070 **360k**	R437 DNP+	R071 DNP
IN11	R072 **270k**	R073 DNP	R074 DNP	R075 DNP	R076 DNP	R077 **330k**	R438 DNP+	R439 DNP+
IN12	R078 **360k**	R079 DNP	R080 **750k**	R440 DNP+	R441 DNP+	R081 **1.1M**	R442 DNP+	R082 **220k**
IN13	R083 **390k**	R084 DNP	R443 DNP+	R085 DNP	R444 DNP+	R086 **270k**	R087 DNP	R088 DNP
IN14	R089 **330k**	R090 DNP	R091 DNP	R092 DNP	R445 DNP+	R093 **300k**	R446 DNP+	R094 DNP
IN15	R447 DNP+	R095 **220k**	R448 DNP+	R096 DNP	R449 DNP+	R097 **120k**	R450 DNP+	R098 DNP
IN16	R099 **330k**	R100 DNP	R101 DNP	R102 DNP	R103 **75k**	R104 DNP	R105 DNP	R106 DNP
IN17	R107 **510k**	R108 DNP	R109 DNP	R110 DNP	R111 **220k**	R112 DNP	R113 DNP	R114 DNP
IN18	R115 **1M**	R116 DNP	R451 DNP+	R117 DNP	R118 DNP	R119 **750k**	R120 DNP	R121 DNP
IN19	R122 **130k**	R452 DNP+	R123 DNP	R124 DNP	R125 DNP	R126 DNP	R453 DNP+	R127 DNP
IN20	R128 **110k**	R454 DNP+	R455 DNP+	R129 DNP	R130 DNP	R131 **820k**	R456 DNP+	R132 DNP
IN21	R457 DNP+	R133 DNP	R458 DNP+	R134 DNP	R135 DNP	R136 **2M**	R459 DNP+	R137 DNP
IN22	R460 DNP+	R138 **1.5M**	R461 DNP+	R139 DNP	R140 DNP	R141 DNP	R462 DNP+	R142 DNP
IN23	R143 **510k**	R144 DNP	R463 DNP+	R145 DNP	R146 **51k**	R147 DNP	R464 DNP+	R148 DNP
IN24	R465 DNP+	R149 **220k**	R150 DNP	R151 DNP	R152 DNP	R153 **150k**	R154 DNP	R155 DNP
IN25	R156 DNP	R157 **390k**	R158 DNP	R159 DNP	R160 DNP	R161 **160k**	R162 DNP	R163 DNP
IN26	R164 DNP	R165 **360k**	R466 DNP+	R166 DNP	R467 DNP+	R167 **150k**	R168 DNP	R169 DNP
IN27	R170 DNP	R171 **360k**	R468 DNP+	R172 DNP	R469 DNP+	R173 **1.3M**	R174 DNP	R175 DNP
IN28	R176 **560k**	R177 DNP	R470 DNP+	R178 DNP	R179 DNP	R180 **430k**	R181 DNP	R471 DNP+
IN29	R182 **330k**	R183 DNP	R472 DNP+	R184 DNP	R185 DNP	R186 **470k**	R187 DNP	R473 DNP+
IN30	R474 DNP+	R188 DNP	R189 DNP	R190 DNP	R191 DNP	R192 DNP	R193 DNP	R475 DNP+
IN31	R194 **240k**	R195 DNP	R196 DNP	R197 DNP	R198 **39k**	R476 DNP+	R199 DNP	R477 DNP+

DC1d · weights IN32–IN63 (right bank) · neurons N12–N15

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). **DNP+** on purple = future-proof site added in the 532-site layout, open on all four cards.

IN	POS N12 (POS_N0)	INV N12 (INV_N0)	POS N13 (POS_N1)	INV N13 (INV_N1)	POS N14 (POS_N2)	INV N14 (INV_N2)	POS N15 (POS_N3)	INV N15 (INV_N3)
IN32	R200 **270k**	R478 DNP+	R201 DNP	R202 DNP	R203 DNP	R204 **470k**	R205 DNP	R206 DNP
IN33	R207 DNP	R208 **330k**	R209 DNP	R479 DNP+	R210 DNP	R211 **160k**	R212 DNP	R480 DNP+
IN34	R213 DNP	R214 **220k**	R215 DNP	R481 DNP+	R216 DNP	R217 **300k**	R218 DNP	R219 DNP
IN35	R220 DNP	R221 **180k**	R222 **130k**	R223 DNP	R224 DNP	R225 **360k**	R482 DNP+	R226 DNP
IN36	R227 DNP	R228 **330k**	R483 DNP+	R229 DNP	R230 DNP	R231 **160k**	R232 **160k**	R233 DNP
IN37	R234 **430k**	R235 DNP	R236 DNP	R237 DNP	R238 DNP	R239 **270k**	R484 DNP+	R240 DNP
IN38	R241 **220k**	R485 DNP+	R242 DNP	R243 DNP	R244 DNP	R245 **240k**	R246 DNP	R486 DNP+
IN39	R247 DNP	R248 **130k**	R249 DNP	R250 DNP	R251 **91k**	R252 DNP	R253 DNP	R254 DNP
IN40	R255 DNP	R256 **430k**	R257 DNP	R487 DNP+	R258 **680k**	R259 DNP	R488 DNP+	R260 DNP
IN41	R261 DNP	R262 **180k**	R263 DNP	R489 DNP+	R264 **160k**	R490 DNP+	R491 DNP+	R265 DNP
IN42	R266 DNP	R267 **470k**	R268 DNP	R492 DNP+	R269 **270k**	R493 DNP+	R270 DNP	R494 DNP+
IN43	R271 DNP	R272 **2.2M**	R273 DNP	R495 DNP+	R274 DNP	R275 **910k**	R496 DNP+	R276 DNP
IN44	R277 DNP	R278 **430k**	R279 DNP	R497 DNP+	R280 DNP	R281 **560k**	R498 DNP+	R282 DNP
IN45	R283 DNP	R284 **430k**	R285 DNP	R286 **180k**	R287 DNP	R288 **1.3M**	R499 DNP+	R289 DNP
IN46	R290 DNP	R500 DNP+	R291 DNP	R501 DNP+	R292 DNP	R293 DNP	R294 DNP	R295 DNP
IN47	R296 **120k**	R297 DNP	R298 DNP	R299 DNP	R300 **100k**	R301 DNP	R302 DNP	R303 DNP
IN48	R304 DNP	R305 **130k**	R502 DNP+	R306 DNP	R307 **150k**	R308 DNP	R309 DNP	R503 DNP+
IN49	R310 DNP	R311 **270k**	R312 DNP	R504 DNP+	R313 **68k**	R505 DNP+	R506 DNP+	R314 DNP
IN50	R315 DNP	R316 **300k**	R507 DNP+	R508 DNP+	R317 **82k**	R318 DNP	R509 DNP+	R319 **240k**
IN51	R320 DNP	R321 **200k**	R322 DNP	R323 DNP	R324 **62k**	R325 DNP	R326 DNP	R327 DNP
IN52	R328 DNP	R329 **150k**	R330 DNP	R331 **330k**	R332 **75k**	R510 DNP+	R333 **160k**	R334 DNP
IN53	R335 DNP	R336 **120k**	R337 DNP	R338 DNP	R339 **62k**	R340 DNP	R511 DNP+	R341 DNP
IN54	R512 DNP+	R342 **75k**	R513 DNP+	R343 DNP	R344 **68k**	R345 DNP	R346 DNP	R514 DNP+
IN55	R347 DNP	R348 **100k**	R349 DNP	R350 DNP	R351 **240k**	R352 DNP	R353 DNP	R354 DNP
IN56	R515 DNP+	R355 DNP	R356 DNP	R357 **200k**	R358 DNP	R359 **36k**	R516 DNP+	R360 DNP
IN57	R361 DNP	R362 **160k**	R363 DNP	R364 DNP	R365 DNP	R366 **82k**	R517 DNP+	R367 DNP
IN58	R518 DNP+	R368 **160k**	R369 DNP	R370 DNP	R519 DNP+	R371 **430k**	R372 DNP	R373 DNP
IN59	R520 DNP+	R374 **150k**	R375 **180k**	R521 DNP+	R522 DNP+	R376 DNP	R377 DNP	R378 DNP
IN60	R523 DNP+	R379 **1.2M**	R380 DNP	R381 DNP	R382 **180k**	R383 DNP	R384 DNP	R385 DNP
IN61	R386 DNP	R387 **160k**	R388 DNP	R389 DNP	R390 DNP	R391 DNP	R524 DNP+	R392 DNP
IN62	R525 DNP+	R393 **200k**	R394 DNP	R395 **200k**	R396 **330k**	R526 DNP+	R527 DNP+	R397 DNP
IN63	R528 DNP+	R398 **100k**	R399 DNP	R400 **390k**	R401 DNP	R402 **160k**	R403 DNP	R404 DNP

DC1d · central resistors (feedback, bias, balance)

x = 47 / 52.6 mm, y = 44–94 mm, grouped by net pair. Balance and feedback are 1% (blue); bias is 5%. Red and purple rows stay open.

Ref	X mm	Y mm	Pad 1	Pad 2	Role	Neuron	Value	Tol	Original ref
R529	47	44	+5V_REF	POS_N0	bias	N12	DNP+ (new)		
R405	47	59	+5V_REF	INV_N0	bias	N12	**750k**	5%	Rb_N12
R530	52.6	44	+5V_REF	POS_N1	bias	N13	DNP+ (new)		
R406	52.6	59	+5V_REF	INV_N1	bias	N13	DNP (open)		
R409	47	69	+5V_REF	POS_N2	bias	N14	DNP (open)		
R407	47	64	+5V_REF	INV_N2	bias	N14	**200k**	5%	Rb_N14
R531	47	49	+5V_REF	POS_N3	bias	N15	DNP+ (new)		
R408	52.6	64	+5V_REF	INV_N3	bias	N15	DNP (open)		
R413	47	79	GND	POS_N0	balance	N12	**15.4k**	1%	R_BAL_1_12
R532	52.6	49	GND	INV_N0	balance	N12	DNP+ (new)		
R414	52.6	79	GND	POS_N1	balance	N13	**52.3k**	1%	R_BAL_1_13
R410	52.6	69	GND	INV_N1	balance	N13	DNP (open)		
R415	47	84	GND	POS_N2	balance	N14	DNP (open)		
R411	47	74	GND	INV_N2	balance	N14	**113k**	1%	R_BAL_1_14
R416	52.6	84	GND	POS_N3	balance	N15	**61.9k**	1%	R_BAL_1_15
R412	52.6	74	GND	INV_N3	balance	N15	DNP (open)		
R417	47	89	INV_N0	V_SUM_0	feedback	N12	**100k**	1%	Rf_N12
R418	52.6	89	INV_N1	V_SUM_1	feedback	N13	**100k**	1%	Rf_N13
R419	47	94	INV_N2	V_SUM_2	feedback	N14	**100k**	1%	Rf_N14
R420	52.6	94	INV_N3	V_SUM_3	feedback	N15	**100k**	1%	Rf_N15

Also fit the fixed parts: U1 TL074, C_U1P and C_U1N 0.1 µF 50 V 0805, J_STRADDLE_EDGE.